

The Fundamental Theorem of Calculus, Part 2

Today's Goals

- ◆ You should be able to explain why two antiderivatives of the same function f differ by a constant.
- ◆ You should be able to use part 2 of the Fundamental Theorem of Calculus to calculate definite integrals.
- ◆ You should be able to find antiderivatives of basic functions.
- ◆ You should be able to distinguish between definite integrals, indefinite integrals, and antiderivatives.

1. (Blue Bell revisited) At the Blue Bell ice cream factory, milk is stored in a 40,000 gallon tank. The rate of milk flowing into and out of the tank t hours after noon one day is $f(t) = 1000 - 300t^2$ gallons per hour. At 2 pm, there are 15,000 gallons of milk in the tank.

(a) Express the net change in the amount of milk between 1 pm and 3 pm as a definite integral.

(b) Let $M(t)$ be the amount of milk in the tank at time t . What is $M'(t)$? What is $M(2)$? Using these two pieces of information, can you find a formula for $M(t)$?

(c) Use your answer to (b) to find the net change in the amount of milk in the tank between 1 pm and 3 pm.

The Fundamental Theorem of Calculus, Part 2 (FTOC 2)

If f is a continuous function on $[a, b]$ and F is any antiderivative of f (that is, F is any function such that $F' = f$), then

$$\int_a^b f(t) dt = \underline{\hspace{2cm}}.$$

2. Evaluate $\int_{-2}^2 4x^3 dx$.

3. Evaluate $\int_4^3 x dx$.

Indefinite Integrals

We use the notation $\int f(x) dx$ to mean “all antiderivatives of f ” or “the most general antiderivative of $f(x)$ ” and call this the indefinite integral of $f(x)$. While the definite integral $\int_a^b f(x) dx$ is a number, the indefinite integral $\int f(x) dx$ is a family of functions.

4. Find each of the following.

(a) $\int (4x^2 + 2) dx$

(b) $\int u\sqrt{u} du$

5. First decide whether each of the following definite integrals should be positive or negative; then evaluate them. Simplify your answers.

(a) $\int_1^0 (4x^2 + 2) dx$

(b) $\int_4^{\pi^2} u\sqrt{u} du$

6. A bronze statue sits in a courtyard. One sunny day, the statue's temperature at noon is 40 degrees Celsius. Suppose that the statue's temperature is changing at a rate of $r(t) = \frac{1}{e^t} + 2$ degrees Celsius per hour, where t is the numbers of hours since noon.

(a) At what times is the statue heating up? At what times is the statue cooling down?

(b) What is the statue's temperature at 1:00 pm?

(c) Do a consistency check: given that the statue's temperature at noon was 40 degrees, do your answers make sense with each other?

7. (a) Which of the following are antiderivatives of $f(x) = \frac{1}{\sqrt{1-x^2}}$?

$\arcsin x$

$\arccos x$

$\arctan x$

$\arcsin x + 7$

$-\arccos x$

- (b) Since $f(x) = \frac{1}{\sqrt{1-x^2}}$ is continuous on its domain $(-1, 1)$, if $F(x)$ is one antiderivative of $f(x)$, then the most general antiderivative of $f(x)$ is $F(x) + C$. Explain why this does *not* contradict your answers to (a).

8. Let $G(x) = \int_{-e^{2x}}^{e^{2x}} \cos(t^2) dt$. Find $G'(x)$.

Hint: This can be done without finding an antiderivative of $\cos(t^2)$.

9. True or false:

(a) If $\int_0^x f(t) dt = \int_0^x g(t) dt$ for all x , then $f(x) = g(x)$.

(b) If $f'(x) = g'(x)$ for all x , then $f(x) = g(x)$.

(c) If $f(x)$ is an increasing function on $(-\infty, \infty)$, then $\int_3^x f(t) dt$ is increasing on $(3, \infty)$.